



EcoGen Solutions
Clean power. Less waste. Smarter farms.

Transforming waste management challenges into renewable energy opportunities for pig farms.

Client Profile: Mid-Sized Piggery



WHO ?

- A privately operated, family-owned farrow-to-finish piggery
- Located in regional Victoria, a key pork-producing region
- Operates around 500 sows, producing pigs to ~99kg live weight

WHAT ?

- Generates ~2,000 tones of effluent annually
- Proposed system includes:
 - Covered anaerobic digester
 - 50 - 100 kW biogas generator
 - Digestate storage for third-party sale

WHY ?

- Reflects a typical Australian mid-sized piggery
- Based on real case data used in industry studies (e.g., APL)
- Ideal scale for biogas: consistent waste, strong ROI, and clear emissions savings

What Problem Are We Solving?

Methane Emissions

Uncovered lagoons release methane – the largest on-farm emissions source (up to 66%).

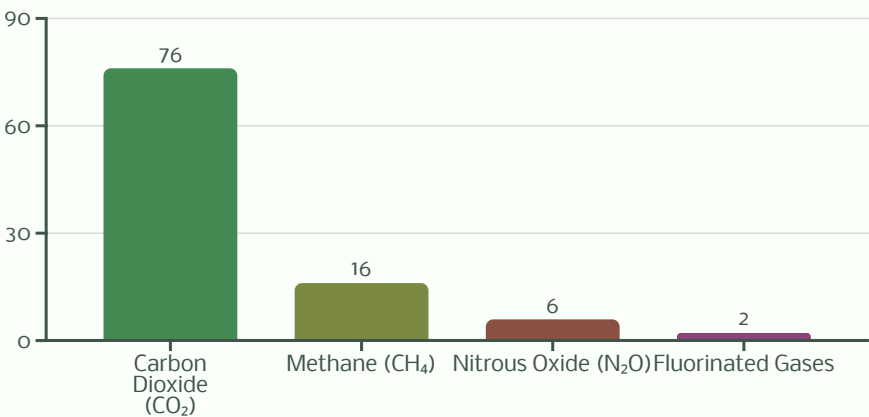
Rising Costs

High and volatile LPG + grid electricity prices reduce farm profitability.

Untapped Value

Nutrient-rich digestate is currently unmanaged or underutilised.

Research Insights



- Agriculture
- Fossil Fuels (Oil & Gas)
- Waste Management
- Other Sources



Emission Sources

75% of piggery emissions come from manure lagoons.

Digestate Value

Worth \$25k - 55k/year as fertilizer.

Energy Potential

More than 100,000 kWh/year possible with biogas.

Aligning With Global Goals

SDG 7 - Affordable & Clean Energy

Energy independence through
biogas



SDG 12 - Responsible Consumption & Production

Waste becomes fertiliser

SDG 13 - Climate Action

Methane reduction = climate
impact

Reframing the Problem

How might we help small-to-medium pig farms reduce carbon emissions and generate revenue using a sustainable waste-to-energy solution?



Identify

Methane emission sources



Convert

Process waste into energy and digestate with anaerobic digestion.



Generate

Deliver clean energy and fertiliser income from a circular system.



System Overview



Manure

Collection from piggery effluent system



Digester

Manure undergoes anaerobic decomposition to produce biogas



Biogas

Biogas collected in a gas holder and piped to the genset for use.



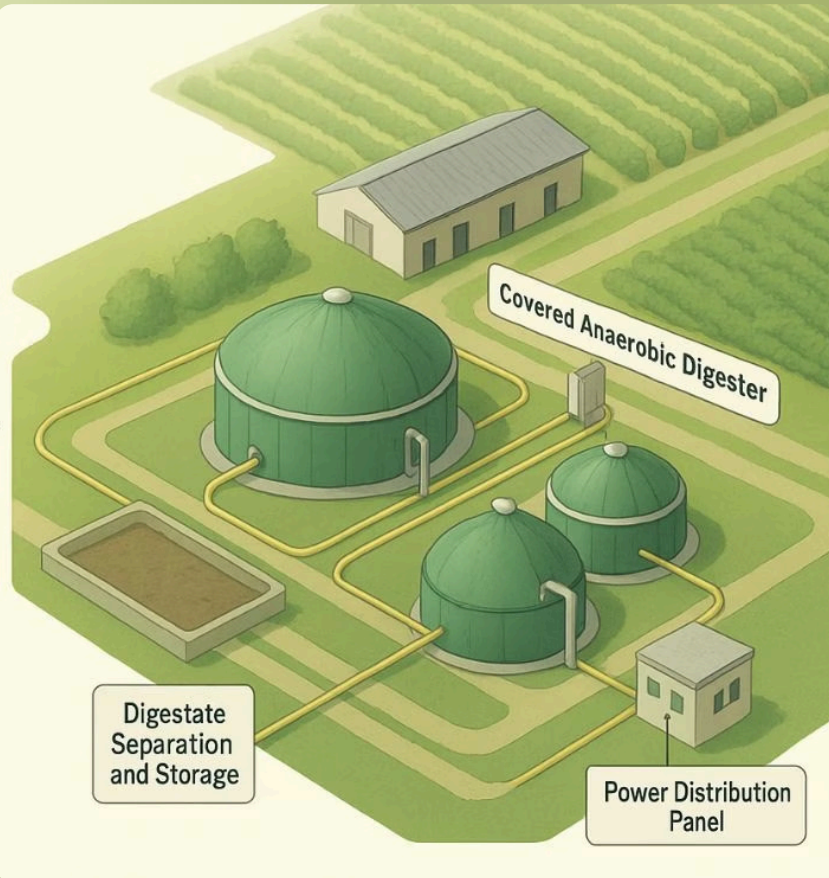
Electricity

On-site electricity from 50-100 kW biogas generator

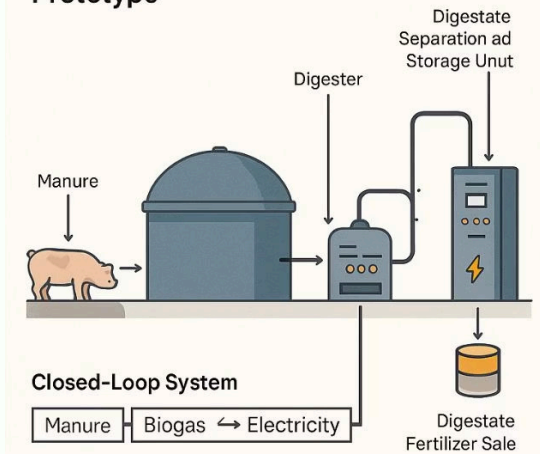


Digestate

Separated and stored for sale (100% sold to third-party buyer)



Prototype



Environmental & Economical Impact

Environmental Impact

70 - 80 %

Decrease in Methane emissions

75%

Emissions Reduction

Potential decrease in farm's carbon footprint

100%

Entire energy output is derived from on-farm waste (manure), replacing fossil fuels

Economical Impact

\$25K–55K/year

Revenue from digestate sales

\$30K/year

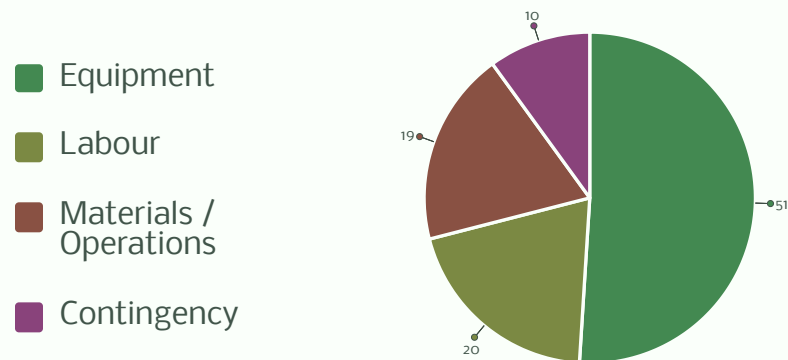
Savings on electricity costs

≈ 9 years

Estimated Payback Period
at \$85K annual benefit

Budget & Timeline

Project Budget Breakdown (%)



Breakdown Summary

Total Project Cost: AUD \$801,295

- **Equipment (51%)** - Digester system, biogas generator, piping, control unit
- **Labour (20%)** - Biogas contractor, electrician, plumber, installation crew
- **Materials/Operations (19%)** - Concrete, fencing, wiring, transport, civil works
- **Contingency (10%)** - Unforeseen costs, price changes, project delays

Project Implementation Schedule

May 2024 - June 2024 Planning & Design	July 2024 - November 2024 Procurement & Construction	Nov 2024 - March 2025 Commissioning & Launch
Feasibility Study Completed	Components Procured	System Testing Begins
Design Finalised	Site Preparation Begins	Staff Training Completed
Permits Submitted	Tank & Equipment Installed	Product launch
	Electrical & Plumbing Works	Final Report & Handover

Team Contribution Breakdown

Team Member	Primary Contributions
Saad	Lead presenter & designer of client persona, research problem, and initial slide structure.
Manav	Primary research analyst - developed insights, charts, UNSDG alignment, and methane context visuals.
Krishna	Reframed problem statement, drafted "How Might We", and supported the prototype diagram creation.
Ali	Detailed system prototype explanation, environmental & economic impact cards, and system flow visuals.
Amal	Budget breakdown, simplified Gantt chart, calculated payback periods, and structured PMP summary slides.